1. Let $T$ be the set of transient states of a Markov chain and let $i, j \in T$. If $\mathbf{S}$ denotes the matrix whose entries $s_{ij}$ are the expected number of time periods that the Markov chain is at state $j$ given it started at state $i$, then it was shown in class that

$$(I - \mathbf{P}_T)\mathbf{S} = I,$$

where $\mathbf{P}_T$ denotes the transition matrix involving the transient states only. Show that $I - \mathbf{P}_T$ is indeed invertible so that

$$\mathbf{S} = (I - \mathbf{P}_T)^{-1}.$$

2. Let

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.4 & 0 & 0.6 & 0 \\ 0 & 0.7 & 0 & 0.3 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Compute $s_{11}, s_{12}, s_{21}, s_{22}, f_{11}, f_{12}, f_{21}, f_{22}$.

3. In the branching processes example of Section 4.7, show that when $\mu = 1$, $\Pi_0 = 1$.

4. (# 64 in book) Consider a branching process having $X_0 = 1$ and $\mu > 1$. Show that if $X_0 = 1$, then the expected number of individuals that ever exist in this population is given by $1/(1-\mu)$. What if $X_0 = n$?

5. (# 66 in book) For a branching process, calculate $\Pi_0$ when

   (a) $P_0 = \frac{1}{4}, P_2 = \frac{3}{4}$.

   (b) $P_0 = \frac{1}{4}, P_1 = \frac{1}{2}, P_2 = \frac{1}{4}$.

   (c) $P_0 = \frac{1}{6}, P_1 = \frac{1}{2}, P_3 = \frac{1}{3}$.